

Pratham Udeshi

AI Engineer · Full Stack Developer · GenAI & Agentic Systems

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PROFILE SUMMARY

Final-year B.Tech Computer Engineering student at K.J. Somaiya (CGPA 8.9) with 6+ internships spanning Agentic AI, LLM systems, full-stack development, and cybersecurity. Designed and shipped production-grade AI agents at Godrej Enterprises, including an Email Agent with human-in-the-loop and an Engineering Specs sheet Analyzer achieving 98% accuracy. Recognized at national hackathons including IIT-Bombay's LLM Hackathon (Top 10) and Crackathon (Finalist). Passionate about building high-impact, scalable technology at the intersection of AI and software engineering.

EDUCATION

K.J. Somaiya School of Engineering

B.Tech Computer Engineering with Honours in Cyber Security

Mumbai, India

2022 – Jun 2026 · CGPA: 8.9

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C/C++, SQL

Web & Full Stack: React, Node.js, Django, Next.js, HTML, CSS, Flutter

AI / GenAI: LLMs, Agentic AI, RAG, Prompt Engineering, LangChain, Semantic Filtering, LLM-as-a-Critic, Computer Vision

Tools & Platforms: Git, REST APIs, Figma, Three.js, CI/CD, Streaming Pipelines, SharePoint / OneDrive APIs

Domains: Cybersecurity, DevOps, FinTech, EdTech, Aerospace, Supply Chain

EXPERIENCE

Godrej Enterprises — AI Engineer Intern · Mumbai

Jan 2026 – Jun 2026

- Engineered a production-grade OneDrive & SharePoint Connector for Godrej's internal Amethyst AI platform, supporting safe read and write operations with multi-layered guardrails to prevent accidental data mishaps.
- Built an intelligent Email Agent capable of listening to all incoming emails, generating context-aware replies using a knowledge base, applying guardrails, and routing to a smart human-in-the-loop when needed — the human can write, escalate to a manager, or acknowledge the sender, and the agent handles execution autonomously.
- Developed the Multi-Address Generation Agent for Godrej Interio, automating an address generation workflow that previously required 2 man-days, completing it in minutes with no manual intervention.
- Built an Engineering Specs sheet Analyzer for an aerospace client: reads and parses complex engineering spec documents, identifies all required materials and manufacturing processes with 98% accuracy, generates a complete Bill of Materials, and flags which processes can be handled in-house vs. outsourced.
- Conducted deep research on drone-based inspection using AI — evaluated programmable drones with modular sensor configurations for industrial inspection use cases.
- Developed a 2D/3D Drawing Analyzer capable of deeply parsing STP-format engineering drawings for detailed analysis, identifying design patterns, and surfacing potential improvements.
- Worked on a Computer Vision use case for a large warehousing facility (Sai Dhaara, Bhiwandi) — safety monitoring for on-ground workers and rack space occupancy detection.
- Core contributor to the official launch of Amethyst, Godrej's Internal AI Platform — recognized in multiple LinkedIn posts by organizational leadership.

Godrej Enterprises — Data Science Intern · Mumbai

Jun 2025 – Jul 2025

- Designed a full LLM security guardrails framework from scratch, capable of blocking sophisticated jailbreak attempts that bypass GPT's default content filters.
- Implemented Regex-based pattern matching, Semantic Filtering, and LLM-as-a-critic as a multi-layered defense system to make guardrails robust across diverse attack vectors.

- Built a comprehensive analytics and reporting dashboard to monitor guardrail performance in real time, identify emerging vulnerabilities, and drive iterative safety improvements.
- Benchmarked system against industry LLM safety standards; collaborated closely with domain experts across legal, operations, and tech to ensure cross-functional compliance.

DeepCytes — *Project Leader* · Remote

Mar 2025 – May 2025

- Engineered a real-time global cyber threat visualization system capable of processing 10,000+ data points per second using optimized streaming pipelines.
- Designed and built a scalable backend that continuously ingests and processes thousands of threat data entries per minute, applying clustering techniques to intelligently group and stream data to the frontend.
- Built a responsive, information-dense frontend that renders live cyberattack data across a global map — showing attack origins, targets, and threat types in real time.

Expert SCM — *Project Leader* · Remote

Oct 2024 – Dec 2024

- Built a dynamic supply chain management simulation game using real-world logistics logic, animations, and interactive decision-making to help students learn SCM concepts.
- Bridged the gap between technical implementation and supply chain theory, making complex operational concepts intuitive and engaging for non-technical learners.

Coding Judge — *Frontend Developer Intern* · Remote

Jul 2024 – Aug 2024

- Implemented lazy-loading, debouncing, and memoization to significantly improve frontend performance, preventing crashes and reducing perceived latency to near-instant levels under heavy usage.
- Delivered under tight deadlines in a high-pressure environment, mentored directly by Microsoft SDEs and IIT-B engineers.

K.J. Somaiya College of Engineering — *Full Stack Developer & Project Lead* — *ProCompiler* · Mumbai Dec 2023 – Aug 2024

- Designed and built ProCompiler, a secure multi-language online compiler (Python, C/C++, Java) used to conduct tamper-proof coding examinations for 5,000+ students.
- Rewrote the system architecture twice to optimize execution performance, improve sandboxing, and add new language support — each iteration significantly improving usability and reliability.
- Led a small dev team, managing sprint planning, code reviews, and deployment cycles from inception to production release.

K.J. Somaiya College of Engineering — *Frontend Developer* — *ProctConnect* · Mumbai

Jul 2023 – Jul 2024

- Developed a centralized academic portal for 5,000+ users enabling seamless communication between proctors, students, and faculty for monitoring, grievances, and academic tracking.
- Scaled backend infrastructure to support live academic monitoring sessions, ensuring reliability during high-concurrency usage.

PROJECTS

IGNITE-WEB | EdTech · AI · Prompt Engineering · github.com/prathamudeshi

- Built a platform that converts natural language prompts into production-ready, editable websites using structured prompt-to-code generation pipelines.
- Added live runtime editing support — users can modify layout, text, and styling in-browser without touching code, bridging prompt generation with design control.
- Focused on AI prompt structuring to reliably map conversational intent to valid web layout logic, minimizing hallucinations in code output.

HOSTKAR | DevOps · CI/CD · Vercel Clone

- Developed a CI/CD web deployment platform where users can deploy static and dynamic sites by simply pasting a GitHub repository link — no manual configuration needed.
- Built with Node.js and Express; handles automatic build triggering, artifact hosting, and route management — replicating core Vercel functionality from scratch.

PROMPT-SEARCH | AI · LLM · Search

- Solved the LLM hallucination problem for search-based queries using auto-enhanced query generation that rewrites user prompts for improved retrieval precision.
- Integrated Google Custom Search Engine (CSE) to ground responses in real, indexed sources rather than model memory.

STOCK-SCOPE | FinTech · Data Visualization

- Built a stock analytics dashboard providing detailed statistics, live charts, and portfolio insights with real-time data visualization.

ROOMERS | Social Media · Real-Time Systems

- Developed a real-time chat application with AJAX-based messaging, emoji/GIF support, and a scalable message delivery architecture.

Flight Booking Frontend | Freelance · Travel Tech

- Built the complete frontend for a client's flight booking system — search, filter, and booking flows with a responsive, user-friendly interface.

Web Data Extractor | Freelance · Automation

- Built an automated data extraction tool for 99acres.com, scraping and structuring property listings into structured, queryable formats.

AWARDS & RECOGNITION

- Top 10 — LLM Hackathon at IIT-Bombay (National-level AI competition)
- Finalist — Crackathon (National-level coding competition)
- 1st Place — Tech Competition organized by ISTE (Indian Society for Technical Education), KJSCE
- 1st Place — Figma Design Competition by Friends of Figma (FOF) Mumbai
- 2nd Place — State-level Mathematics Competition
- Featured in official Godrej LinkedIn media posts for contributions to the Amethyst AI Platform launch

CERTIFICATIONS

- Generative AI with Large Language Models — DeepLearning.AI / Coursera
- Prompt Engineering for ChatGPT — Coursera
- Learn to Program: The Fundamentals — University of Toronto / Coursera

EXTRACURRICULAR & INTERESTS

- Active hackathon participant — regularly competes in national and institute-level competitions in AI, design, and coding.
- Sports: Squash, Table Tennis (3rd Place — Office Table Tennis Tournament at Godrej), Cycling.
- Creative pursuits: Video editing, singing, acting.
- Passionate about building high-scalability tech solutions that bridge the gap between AI research and real-world deployment.